

REVIEW OF RELATED LITERATURE

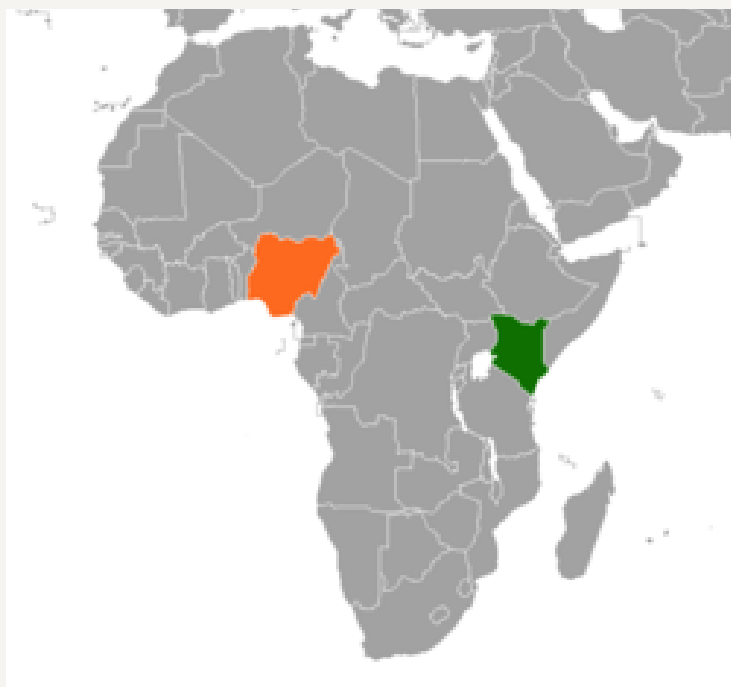
Data-Driven Property Valuation for Cebu City

Presented By

CHRIS DOMINIC ESTREBA | FEB 2026

“Data scarcity — not appraiser incompetence — is the core problem”

"The core reason... is **limited information** and not valuer misconduct."
— Cheloti & Mooya (2021)



Finding	Source
" Limited information " = #1 problem (Mean Rank 2.91)	Kenya
92.7% cite insufficient market evidence	Lagos
Error: +24% to +51% vs $\pm 10\%$ global norm	Lagos

The Philippines faces the same structural gap

Phil. — Ramolete et al., 2023



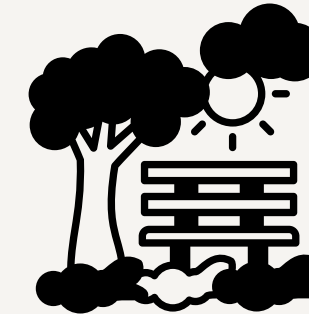
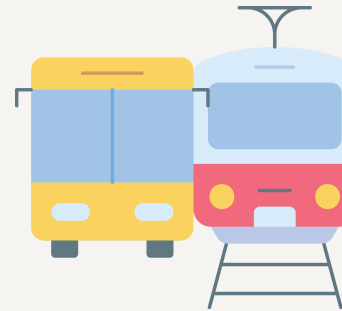
- **Fragmentation:** Different agencies (BIR, LGU, LRA) use different methods, timelines, and standards
- **Political intervention:** Keeps valuations artificially low
- **Data poverty:** Transaction records exist but are not processed or tabulated
- **Heuristics overload:** Valuers rely on mental shortcuts due to missing data

Problem	Detail
Outdated Zonal Values	Only 60% of LGUs updated (2017–2020)
Missing Market schedules	Only 37% submitted SMVs
Mutiple valuation systems	2–3 different values for same property
Agency Disconnect	BIR vs LGU vs LRA with overlapping mandates



The only Cebu-specific empirical study on land value determinants

Agosto Study



Top 5 Determinants (by Factor Loading)

1. Accessibility to public transportation
2. Recreational facilities
3. Open spaces and parks
4. Environmental quality
5. Level of ownership

Aspect	Detial
Sample	51 practitioners
Variables	31 tested → 11 factors (PCA)
Limitation	Residential properties only

Tree-based models outperform neural networks on small data

Tanzania — Nyanda et al., 2024

Model	MAPE
Neural Network	108.6% ❌
Random Forest	52.7%
Boosting	48.0% ✅

- **Sample:** 954 observations (524 formal + 430 informal)
- **Training set:** Only 419 observations (formal market)
- **Features:** Location, structural, temporal, proximity

Sample size guideline:

Neural networks typically need **10–25×** more data than tree-based models to perform well. Tanzania's 419 training samples worked for RF/Boosting but was far too small for NN.

Listing descriptions contain predictive signal

Ottawa — Zhang et al., 2024

Sample:: 10,251 listings

Method: Self-trained Word2Vec + DNN

R²: 0.79

Key Finding:

Self-trained Word2Vec outperformed pre-trained BERT by 44%

Implications

1. Lamudi/DotProperty listing text could provide similar gains
2. Domain-specific training > generic pre-trained models

★ FEATURED ⌚ SHORT-TERM RENTAL

1/13 For Rent | INSIDE CLARK, PAMPANGA | 1 bedroom 1 bathroom 1 guest toilet

Fully-Furnished Studio Condo Unit for Rent located inside Clark Freeport Zone, Pampanga ₱ 35,000/month

📍 Ninoy Aquino Avenue, Clark Center Townhomes, Mabalacat, Pampanga

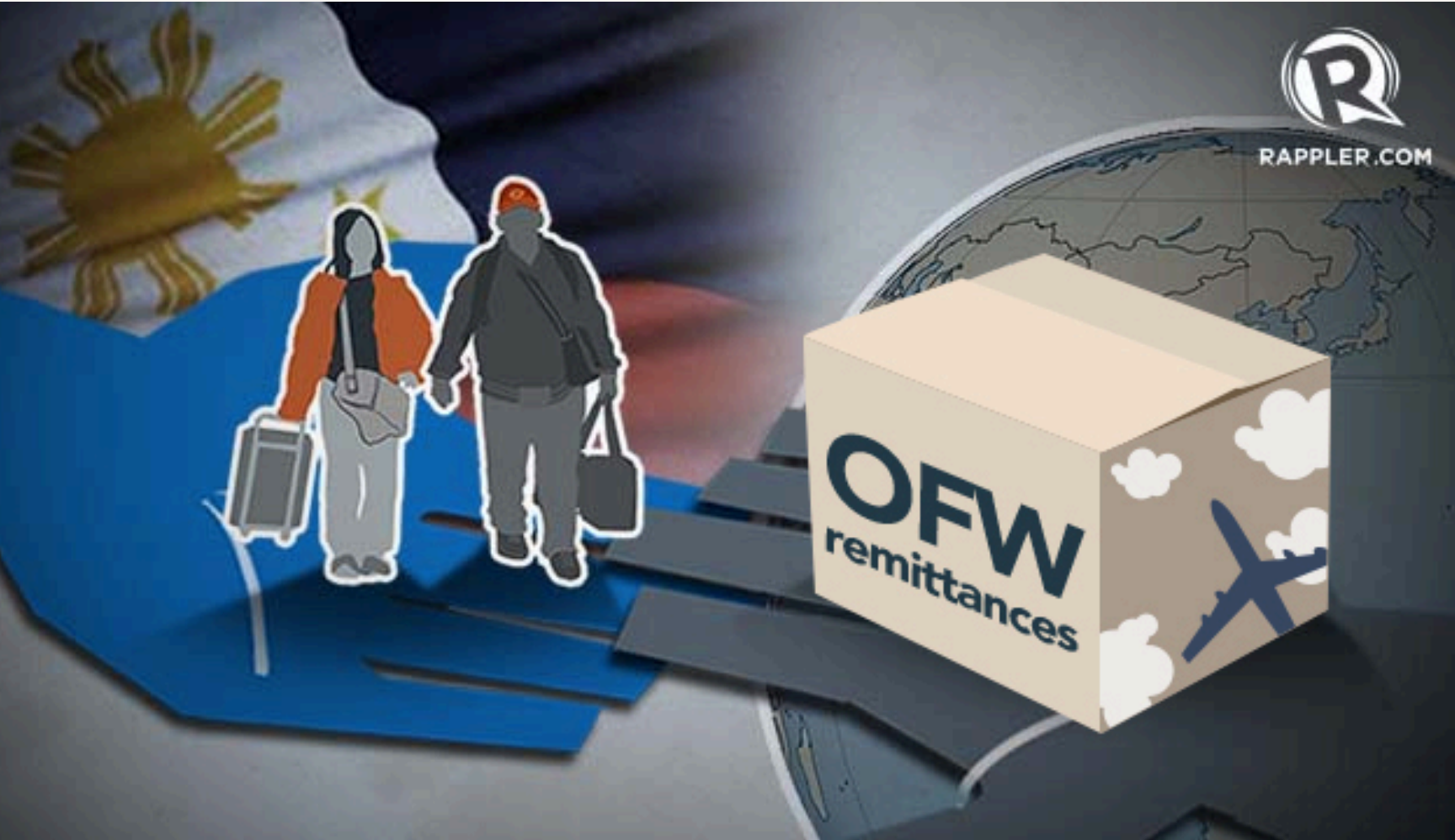
House type: Condo **Offer type:** For Rent
Contract duration: 1 Years **Usable area:** 36 sqm

12 Nov 2025 - Published by Rancel Broñola | Property on Stage

Exchange rates predict property values better than inflation

Nigeria — Nworah et al., 2023

Variable	Correlation
Exchange Rate	$r = 09.25$
Inflation	$r = -0.508$



Cebu Implication:
Peso depreciation → More PHP per USD →
Higher OFW buying power → Real estate
demand ↑

Metric	Value	Source
2024 Remittances	USD 38.3 billion (record high)	BSP
YoY Growth	3.00%	BSP
Central Visayas	"Sustained demand for residential/commercial	NEDA

Models augment — they don't replace — professional judgment

IMF (International Monetary Fund)
House-Prices-at-Risk

Market Type	5th Percentile Risk	Meaning
Emerging Markets	-12%	Worst 5% scenario = 12% price crash
Advanced Economies	-10.50%	Lower downside risk

Thesis Positioning
Our model = Decision Support Tool
with **SHAP transparency**

IVS 2025 (International Valuation Standards) AVM*
Requirements

Requirement	Detail
No standalone compliance	AVM alone cannot produce IVS-compliant valuation
Valuer oversight	Must apply professional judgment on inputs/outputs
Four characteristics	Accuracy, Completeness, Timeliness, Transparency
Documentation	Model selection, inputs, limitations must be justified

*Automated Valuation Model**

The literature provides the toolkit. Cebu is the missing case study.

What Literature Agrees On

1. Data scarcity is THE core problem
2. Tree-based ML (RF/XGBoost) beats NN on small data
3. Text features add 10–44% accuracy
4. Models must support, not replace, humans

Thesis Contribution

- **First Cebu model** using actual transaction data
- **Hybrid:** BDO foreclosures (floor) + Lamudi (ceiling) + broker consultations
- SHAP interpretability for **IVS compliance**

The Gap

No Cebu-specific, transaction-based, ML-augmented model exists.

Approaches Considered but Rejected

Satellite luminosity proxies: Not useful for Grade B–C countries like Philippines (Chen & Nordhaus, 2011)